A4 Browser History Manager

Assignment:

Implement a browser navigation system using your previously developed doubly linked list, iterator, and deque classes. The program must manage visited pages, the current page, forward-history invalidation, range deletion, and a FIFO queue of pending visits

Requirements:

- Reuse Data structures as application components.
- Use bidirectional iterators for navigation, traversal, insertion, and removal
- Preserve valid current-page state during mutations.
- Use a deque to process pending navigation cases requests in FIFO order.
- Test normal operations and edge cases and explain implementation decisions.

Implementation:

I created a class, FoogleHistory with the private members front, rear, and page as pointer variables. The pointer variables front and rear act as place holders for the entire list of history with rear holding the place of the oldest page visited and front holding the place of the newest page in the list. The variable page points to the URL that is currently being viewed. For pending requests, I created a second deque struct, FutureList that is initialized as a private member of FoogleHistory, pending. This includes FutureList in FoogleHistory using composition and allows FoogleHistory to manage both the future requests and the current history.

In addition to the required functions, I also added bool historyEmpty() that returns true if front is equal to nullptr. During the build and debugging process I used assert((front == nullptr) == (rear == nullptr)) inside historyEmpty() to identify and fix cases that created a situation where front == nullptr but rear still pointed to a node.

I also added a function void showAll() const; that will print all nodes from rear to front regardless of the URL that page is currently pointing to. During debugging this was used to test cases where history was supposed to be deleted, specifically after visit when there is a forward history. After visit() is called, all forward history is supposed to be deleted. Using showAll() I was able to verify that page = front and front → next == nullptr.

makefile:

A debug log was created with this program using #include <cassert>.

- make RELEASE=1
  - Compile and execute foogle.exe without debug logs
- make run:
  - compiles and executes foogle.exe
- make clean:
- deletes previous object files and foogle.exe

Required Tests:

Test	Scenario	Expected result
T1	Visit three pages	Chronological order correct
T2	Back twice	Correct current page
T3	Forward once	Correct current page
T4	Back from first	No state change
T5	Forward from last	No state change
T6	Visit after back	Forward entries removed
T7	Delete range excluding current	Current preserved
T8	Delete range containing current	Replacement follows specified rule
T9	Delete all history	Empty state safe
T10	Queue/process three URLs	FIFO order
T11	Process empty queue	Safe false result
T12	Operate after deleting all history	Program remains functional

T1 – Visit three pages: Success

- Test Implementation:
  - Verify empty list
  - Vist page
  - Display current page

```cpp
#include <iostream>
#include "A3List.h"


int main() {
FoogleHistory hx;

    //T1
    std::cout << std::endl << "T1: Visit three pages" << std::endl;
    hx.showCurrentPage(); // list is empty
    hx.visit("www.one.com"); //page one
    hx.showCurrentPage();
    hx.visit("www.two.com"); //page two
    hx.showCurrentPage();
    hx.visit("www.three.com"); //page three
    hx.showCurrentPage();

}

```

- Expected output: Chronological order correct

```
PS C:\Users\zachb\1_Fundamental_Data_Structure\assignments\Assignment_4> make RELEASE=1
g++ -O2 -DNDEBUG -c main.cpp
g++ main.o A3List.o A3Deque.o -o foogle.exe
./foogle.exe

T1: Visit three pages
Empty list, no current page
Current Page: www.one.com
Current Page: www.two.com
Current Page: www.three.com
PS C:\Users\zachb\1_Fundamental_Data_Structure\assignments\Assignment_4>
```

T2 – Back twice: Success

- Test implementation:
  - Verify history, three sites visited
  - Execute back() twice
  - Display current page

```cpp
#include <iostream>
#include "A3List.h"

int main() {
    FoogleHistory hx;

    hx.visit("www.one.com");
    hx.visit("www.two.com");
    hx.visit("www.three.com");
    hx.showHistory();

    //T2
    std::cout << std::endl << "T2: Back twice" << std::endl;
    hx.back();
    hx.back();
    hx.showCurrentPage();

}
```

- Output – Current Page: www.one.com

```
PS C:\Users\zachb\1_Fundamental_Data_Structure\assignments\Assignment_4> make RELEASE=1
g++ -O2 -DNDEBUG -c main.cpp
g++ main.o A3List.o A3Deque.o -o foogle.exe
./foogle.exe
DISPLAYING HISTORY:
Current page: www.three.com
Previous Page: www.two.com
Previous Page: www.one.com
Previous Page: History is empty

T2: Back twice
Current Page: www.one.com
PS C:\Users\zachb\1_Fundamental_Data_Structure\assignments\Assignment_4>
```

T3 – Forward Once: Success

- Implementation:
  - Verify history, three visited sites, using showHistory();
  - Execute back()
  - Verify using showCurrentPage();
  - Execute forward();
  - Verify using showCurrentPage();

```cpp
#include <iostream>
#include "A3List.h"

int main() {
    FoogleHistory hx;

    hx.visit("www.one.com");
    hx.visit("www.two.com");
    hx.visit("www.three.com");
    hx.showHistory();

    //T3
    std::cout << std::endl << "T3: Forward once" << std::endl;
    hx.back();
    hx.showCurrentPage();
    hx.forward();
    hx.showCurrentPage();

}
```

- Output:
  - After back() – Current Page: www.two.com
  - After forward() – Current Page: www.three.com

```
PS C:\Users\zachb\1_Fundamental_Data_Structure\assignments\Assignment_4> make run
g++ -g -c main.cpp
g++ main.o A3List.o A3Deque.o -o foogle.exe
./foogle.exe
DISPLAYING HISTORY:
Current page: www.three.com
Previous Page: www.two.com
Previous Page: www.one.com
Previous Page: History is empty

T3: Forward once
Current Page: www.two.com
Current Page: www.three.com
PS C:\Users\zachb\1_Fundamental_Data_Structure\assignments\Assignment_4>
```

T4 – Back from first: Success

- Implementation:
  - Verify history using showHistory()
  - Execute back();

```cpp
#include <iostream>
#include "A3List.h"

int main() {
    FoogleHistory hx;

    hx.visit("www.one.com");
    hx.showHistory();

    //T4
    std::cout << std::endl << "T4: Back from first" << std::endl;
    hx.back();

}
```

- Output:
  - No previous history
  - “Empty list or no previous history”

```
PS C:\Users\zachb\1_Fundamental_Data_Structure\assignments\Assignment_4> make RELEASE=1
./foogle.exe
DISPLAYING HISTORY:
Current page: www.one.com
Previous Page: History is empty

T4: Back from first
Empty list or no previous history
PS C:\Users\zachb\1_Fundamental_Data_Structure\assignments\Assignment_4>
```

T5 – Forward from last: Success

- Implementation:
  - Verify valid history using showHistory()
  - Execute forward()

```cpp
#include <iostream>
#include "A3List.h"

int main() {
    FoogleHistory hx;

    hx.visit("www.one.com");
    hx.visit("www.two.com");
    hx.visit("www.three.com");
    hx.showHistory();

//T5
    std::cout << std::endl << "T5: Forward from last" << std::endl;
    std::cout << "Forward from last: " << std::endl;
    hx.forward();
}
```

- Output:
  - Forward from last: Empty list or no forward history

```
PS C:\Users\zachb\1_Fundamental_Data_Structure\assignments\Assignment_4> make run
./foogle.exe
DISPLAYING HISTORY:
Current page: www.three.com
Previous Page: www.two.com
Previous Page: www.one.com
Previous Page: History is empty

T5: Forward from last
Forward from last:
Empty list or no forward history
PS C:\Users\zachb\1_Fundamental_Data_Structure\assignments\Assignment_4>
```

T6 – Visit after back: Success

- Implementation:
  - Verify valid history using showHistory();
  - Execute back();
  - Execute visit();
  - Verify forward history deleted using showAll();

```cpp
#include <iostream>
#include "A3List.h"

int main() {
    FoogleHistory hx;

    std::cout << std::endl << "T1:" << std::endl;
    hx.visit("www.one.com");
    hx.visit("www.two.com");
    hx.visit("www.three.com");
    hx.showHistory();

    //T6
    std::cout << "T6: Visit after back: " << std::endl;
    hx.back();
    hx.showCurrentPage();
    hx.visit("wwww.four.com");
    hx.showAll();
}
```

- Output:
  - Current page before test is www.three.com. Back() is executed and current page becomes www.two.com. After visit(www.four.com) current page becomes www.four.com and forward history is deleted which is demonstrated by using showAll(), www.three.com was deleted.

```
PS C:\Users\zachb\1_Fundamental_Data_Structure\assignments\Assignment_4> make run
g++ -g -c main.cpp
g++ main.o A3List.o A3Deque.o -o foogle.exe
./foogle.exe

T1:
DISPLAYING HISTORY:
Current page: www.three.com
Previous Page: www.two.com
Previous Page: www.one.com
Previous Page: History is empty
T6: Visit after back:
Current Page: www.two.com
wwww.four.com
www.two.com
www.one.com
PS C:\Users\zachb\1_Fundamental_Data_Structure\assignments\Assignment_4>
```

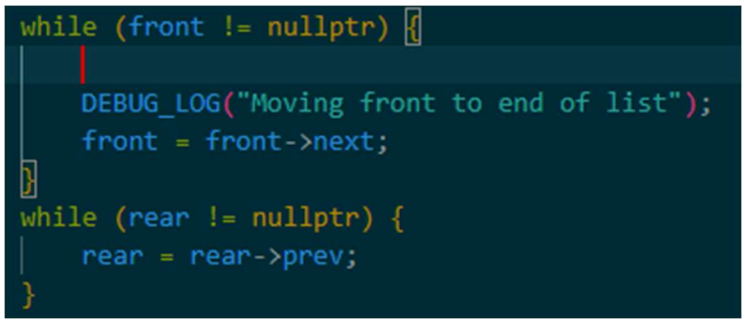
T7 – Delete range excluding current: FAIL

- Implementation
  - Verify history using showHistory();
  - With the last node as the current page execute deleteHistory(0,4)
    - This will delete all but the current page
- Output
  - All history is empty
  - Full debug log:

```
PS C:\Users\zachb\1_Fundamental_Data_Structure\assignments\Assignment_4> make run
g++ -g -c main.cpp
g++ -g -c A3List.cpp
g++ -g -c A3Deque.cpp
g++ main.o A3List.o A3Deque.o -o foogle.exe
./foogle.exe
[debug] FutureList default constructor called
[debug] FoogleHistory default constructor called
[debug] FoogleHistory::vist called
[debug] FoogleHistory::historyEmpty called
[debug] checking for empty history
[debug] FoogleHistory::visit history empty
[debug] FoogleHistory::vist called
[debug] FoogleHistory::historyEmpty called
[debug] checking for empty history
[debug] FoogleHistory::vist deleting forward history
[debug] FoogleHistory::visit create new node, front = new node
[debug] FoogleHistory::vist called
[debug] FoogleHistory::historyEmpty called
[debug] checking for empty history
[debug] FoogleHistory::vist deleting forward history
[debug] FoogleHistory::visit create new node, front = new node
[debug] FoogleHistory::vist called
[debug] FoogleHistory::historyEmpty called
[debug] checking for empty history
[debug] FoogleHistory::vist deleting forward history
[debug] FoogleHistory::visit create new node, front = new node
[debug] FoogleHistory::vist called
[debug] FoogleHistory::historyEmpty called
[debug] checking for empty history
[debug] FoogleHistory::vist deleting forward history
[debug] FoogleHistory::visit create new node, front = new node
[debug] FoogleHistory::vist called
[debug] FoogleHistory::historyEmpty called
[debug] checking for empty history
[debug] FoogleHistory::vist deleting forward history
[debug] FoogleHistory::visit create new node, front = new node
[debug] FoogleHistory::showHistory called
[debug] FoogleHistory::historyEmpty called
[debug] checking for empty history
DISPLAYING HISTORY:
Current page: www.six.com
Previous Page: www.five.com
Previous Page: www.four.com
Previous Page: www.three.com
Previous Page: www.two.com
Previous Page: www.one.com
Previous Page: History is empty
T7: Delete range excluding current
[debug] FoogleHistory::deleteHistory called
[debug] Check for empty list
```

```
[debug] FoogleHistory::historyEmpty called
[debug] checking for empty history
[debug] Validate Range
[debug] Deleting forward history
[debug] Ataching new ends
[debug] Before = nullptr
[debug] Moving front to end of list
[debug] FoogleHistory::showHistory called
[debug] FoogleHistory::historyEmpty called
[debug] checking for empty history
[debug] FoogleHistory::showHistory empty check true, list is empty
List is empty
```

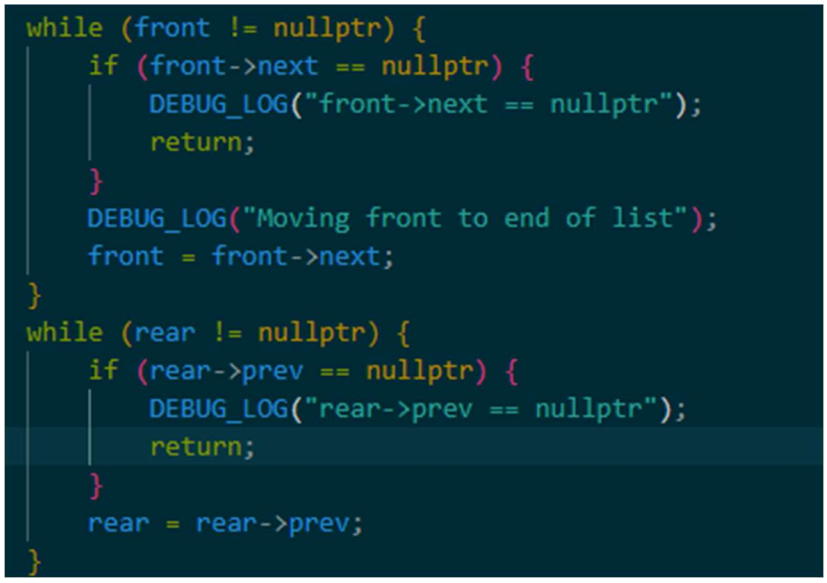
- Bug Report:
    - The bug occurred after checking whether the pointer front was still pointing at the last node. The program incorrectly thought that it was not pointing at the last node and advanced it forward once causing front == nullptr.
    - This screen shot is the applicable line of code:



```
while (front != nullptr) {

    DEBUG_LOG("Moving front to end of list");
    front = front->next;
}
while (rear != nullptr) {
    rear = rear->prev;
}
```

    - If the while loop is changed to while (front->next != nullptr) then the assertion in the historyEmpty() fails and the program crashes.
- Fix:
    - Add additional check before advancing front using an if statement inside the while loop.



```
while (front != nullptr) {
    if (front->next == nullptr) {
        DEBUG_LOG("front->next == nullptr");
        return;
    }
    DEBUG_LOG("Moving front to end of list");
    front = front->next;
}
while (rear != nullptr) {
    if (rear->prev == nullptr) {
        DEBUG_LOG("rear->prev == nullptr");
        return;
    }
    rear = rear->prev;
}
```

T7 – Delete range excluding current: Success

- Implementation
  - Verify history using showHistory();
  - With the last node as the current page execute deleteHistory(0,4)
    - This will delete all but the current page

```cpp
#include <iostream>
#include "A3List.h"

int main() {
    FoogleHistory hx;

    hx.visit("www.one.com");
    hx.visit("www.two.com");
    hx.visit("www.three.com");
    hx.visit("www.four.com");
    hx.visit("www.five.com");
    hx.visit("www.six.com");
    hx.showHistory();

    //T7
    std::cout << "T7: Delete range excluding current" << std::endl;
    hx.deleteHistory(1, 3);
    hx.showHistory();

}
```

- Output:
  - The first, fifth, and sixth pages remain in the list

```
PS C:\Users\zachb\1_Fundamental_Data_Structure\assignments\Assignment_4> make RELEASE=1
g++ -O2 -DNDEBUG -c main.cpp
g++ -O2 -DNDEBUG -c A3List.cpp
g++ -O2 -DNDEBUG -c A3Deque.cpp
g++ main.o A3List.o A3Deque.o -o foogle.exe
./foogle.exe
DISPLAYING HISTORY:
Current page: www.six.com
Previous Page: www.five.com
Previous Page: www.four.com
Previous Page: www.three.com
Previous Page: www.two.com
Previous Page: www.one.com
Previous Page: History is empty
T7: Delete range excluding current
6
DISPLAYING HISTORY:
Current page: www.six.com
Previous Page: www.five.com
Previous Page: www.one.com
Previous Page: History is empty
```

T8 – Delete history including current page: Success

- Implementation:
  - Verify history using showHistory();
  - Move current page using back();
  - With current page on page 3 (www.three.com) execute deleteHistory(1,3);

```cpp
#include <iostream>
#include "A3List.h"

int main() {
    FoogleHistory hx;

    hx.visit("www.one.com");
    hx.visit("www.two.com");
    hx.visit("www.three.com");
    hx.visit("www.four.com");
    hx.visit("www.five.com");
    hx.visit("www.six.com");
    hx.showHistory();

    //T7
    std::cout << "T8: Delete range including current page" << std::endl;
    hx.back();
    hx.back();
    hx.back();
    hx.showCurrentPage();
    std::cout << "There are six pages in the history and the currentpage"
        << "is www.three.com (page 3)\n deleting pages 2-4" << std::endl;
    hx.deleteHistory(1, 3);
    hx.showCurrentPage();
    hx.showAll();
}
```

- Output:
  - Current page moves to the closest point, page 1.
  - showHistory(); will only show history from the current page back, which would be page 1. showAll() was used to list all the pages including the forward history demonstrating that the pages after the delete range are still intact.

```
PS C:\Users\zachb\1_Fundamental_Data_Structure\assignments\Assignment_4> make run
g++ -g -c main.cpp
g++ main.o A3List.o A3Deque.o -o foogle.exe
./foogle.exe
DISPLAYING HISTORY:
Current page: www.six.com
Previous Page: www.five.com
Previous Page: www.four.com
Previous Page: www.three.com
Previous Page: www.two.com
Previous Page: www.one.com
Previous Page: History is empty
T8: Delete range including current page
Current Page: www.three.com
There are six pages in the history and the currentpageis www.three.com (page 3)
 deleting pages 2-4
6
Current Page: www.one.com
www.six.com
www.five.com
www.one.com
PS C:\Users\zachb\1_Fundamental_Data_Structure\assignments\Assignment_4>
```

T9 – Delete all history – Success

- Implementation:
  - Verify valid history is present using showHistory();
  - Execute deleteHistory(0,5) to delete entire history;
  - Execute showCurrentPage() and then showAll(); to demonstrate no current page and no forward or backward history.
  - Execute visit("www.seven.com"); and showCurrentPage(); to demonstrate full functionality after deletion.

```
#include <iostream>
#include "A3List.h"

int main() {
    FoogleHistory hx;

    hx.visit("www.one.com");
    hx.visit("www.two.com");
    hx.visit("www.three.com");
    hx.visit("www.four.com");
    hx.visit("www.five.com");
    hx.visit("www.six.com");
    hx.showHistory();

    //T7
    std::cout << "T9: Delete all history" << std::endl;
    hx.back();
    hx.back();
    hx.back();
    hx.showCurrentPage();
    std::cout << "There are six pages in the history and the currentpage"
        << "is www.three.com (page 3)\n deleting pages 2-4" << std::endl;
    hx.deleteHistory(0, 5);
    hx.showCurrentPage();
    hx.showAll();
    hx.visit("www.seven.com");
    hx.showCurrentPage();
}
```

- Output:
  - showCurrentPage() prints Empty list, no current page
  - showAll() prints All history is empty
  - After visit("www.seven.com"); showCurrentPage() prints CurrentPage: www.seven.com
  - Full functionality after history deletion is demonstrated.

```
PS C:\Users\zachb\1_Fundamental_Data_Structure\assignments\Assignment_4> make run
g++ -g -c main.cpp
g++ main.o A3List.o A3Deque.o -o foogle.exe
./foogle.exe
DISPLAYING HISTORY:
Current page: www.six.com
Previous Page: www.five.com
Previous Page: www.four.com
Previous Page: www.three.com
Previous Page: www.two.com
Previous Page: www.one.com
Previous Page: History is empty
T9: Delete all history
Current Page: www.three.com
There are six pages in the history and the currentpageis www.three.com (page 3)
 deleting pages 2-4
Empty list, no current page
All history is empty
Current Page: www.seven.com
PS C:\Users\zachb\1_Fundamental_Data_Structure\assignments\Assignment_4>
```

T10 – Queue/process three URLs – Success

- Implementation:
  - Show valid history and current page
  - Use enqueueVisit(const string& url) to add 3 pending websites
  - Print pending requests in FIFO order using void showPendingRequests() const;
  - Use bool processNextRequest() three times to execute requests in FIFO order.
  - Execute showPendingRequests() to show deque is empty
  - Execute showHistory() to show all pages have been visited.

```
#include <iostream>
#include "A3List.h"

int main() {
    FoogleHistory hx;

    hx.visit("www.one.com");
    hx.visit("www.two.com");
    hx.visit("www.three.com");
    hx.showHistory();

    //T7
    std::cout << "T10: Queue process three URLS" << std::endl;
    hx.enqueueVisit("www.four.com");
    hx.enqueueVisit("www.five.com");
    hx.enqueueVisit("www.six.com");
    hx.showPendingRequest();
    hx.processNextRequest();
    hx.processNextRequest();
    hx.processNextRequest();
    hx.showPendingRequest();
    hx.showHistory();
}
```

- Output:
  - Pending Requests printed in FIFO order
  - After processNextRequest() was executed three times Pending requests list is empty
  - Display history lists the history in the correct order

```
PS C:\Users\zachb\1_Fundamental_Data_Structure\assignments\Assignment_4> make run
g++ -g -c main.cpp
g++ main.o A3List.o A3Deque.o -o foogle.exe
./foogle.exe
DISPLAYING HISTORY:
Current page: www.three.com
Previous Page: www.two.com
Previous Page: www.one.com
Previous Page: History is empty
T10: Queue process three URLS
Pending Requeusts:
Next URL: www.four.com
Next URL: www.five.com
Next URL: www.six.com
Pending Requeusts:
List is empty
DISPLAYING HISTORY:
Current page: www.six.com
Previous Page: www.five.com
Previous Page: www.four.com
Previous Page: www.three.com
Previous Page: www.two.com
Previous Page: www.one.com
Previous Page: History is empty
PS C:\Users\zachb\1_Fundamental_Data_Structure\assignments\Assignment_4>
```

T11 – Process Empty Queue: Success

- Implementation:
  - Verify valid history and current page using showHistory();
  - Show empty queue using showPendingRequest();
  - Execute processNextRequest();
  - Verify functionality and history using visit("www.four.com"); and showHistory();

```cpp
#include <iostream>
#include "A3List.h"

int main() {
    FoogleHistory hx;

    hx.visit("www.one.com");
    hx.visit("www.two.com");
    hx.visit("www.three.com");
    hx.showHistory();

    //T7
    std::cout << "T11: Process empty queue" << std::endl;
    hx.showPendingRequest();
    hx.processNextRequest();
    hx.visit("www.four.com");
    hx.showHistory();
}
```

- Output:
  - Pending Requests: list empty – no pending requests
  - After processNextRequest(); List is empty
  - Visit() and showHistory() both executed showing full functionality

```
PS C:\Users\zachb\1_Fundamental_Data_Structure\assignments\Assignment_4> make run
g++ -g -c main.cpp
g++ main.o A3List.o A3Deque.o -o foogle.exe
./foogle.exe
DISPLAYING HISTORY:
Current page: www.three.com
Previous Page: www.two.com
Previous Page: www.one.com
Previous Page: History is empty
T11: Process empty queue
Pending Requeusts:
List is empty
List is empty
DISPLAYING HISTORY:
Current page: Www.four.com
Previous Page: www.three.com
Previous Page: www.two.com
Previous Page: www.one.com
Previous Page: History is empty
PS C:\Users\zachb\1_Fundamental_Data_Structure\assignments\Assignment_4>
```

T12 – Operate after deleting all history: Success

- Implementation:
  - Verify valid history and current page using showHistory();
  - Execute deleteHistory(0,2); to delete all three pages
  - Execute showAll(); to show empty list/history
  - Execute visit("www.four.com"); showCurrentPage(); back(); forward();

```
#include <iostream>
#include "A3List.h"

int main() {
    FoogleHistory hx;

    hx.visit("www.one.com");
    hx.visit("www.two.com");
    hx.visit("www.three.com");
    hx.showHistory();

    //T7
    std::cout << "T12: Operate after deleting all history" << std::endl;
    hx.deleteHistory(0,2);
    hx.showAll();
    hx.visit("www.four.com");
    hx.showCurrentPage();
    hx.visit("www.five.com");
    hx.showCurrentPage();
    hx.back();
    hx.showCurrentPage();
    hx.forward();
    hx.showCurrentPage();
}
```

- Output:
  - After showAll(); -- All history is empty
  - Demonstration of visit() and back() and forward() successful.

```
PS C:\Users\zachb\1_Fundamental_Data_Structure\assignments\Assignment_4> make run
g++ -g -c main.cpp
g++ main.o A3List.o A3Deque.o -o foogle.exe
./foogle.exe
DISPLAYING HISTORY:
Current page: www.three.com
Previous Page: www.two.com
Previous Page: www.one.com
Previous Page: History is empty
T12: Operate after deleting all history
All history is empty
Current Page: www.four.com
Empty list or no previous history
Current Page: www.four.com
Empty list or no forward history
Current Page: www.four.com
PS C:\Users\zachb\1_Fundamental_Data_Structure\assignments\Assignment_4>
```